

Daniel Chigbu

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EDUCATION

Loughborough University

Bachelor of Science with Honours in Computer Science

- *Fundamentals of Software Engineering: 77%*

Leicestershire, UK

Sept. 2025 – June 2029

Gravesend Grammar School

Studied Computer Science, Maths, Business — Grades: ABB

Kent, UK

Sept. 2023 – June 2025

EXPERIENCE

JP Morgan Chase Software Engineering

July 2026

Job Simulation

Forage

- Integrated Kafka into a Spring Boot microservice to consume and deserialize high-volume transaction messages using a configurable topic and embedded Kafka test framework.
- Developed a REST endpoint for querying user balances, returning JSON responses through a Spring controller while maintaining clean architectural boundaries.

Quantum Software Engineering

June 2026

Job Simulation

Forage

- Developed an interactive Dash application that enabled the client to assess the impact of price changes on sales and profitability.
- Implemented a test suite to verify the Dash application is working and a bash script to automatically run the test suite.

PROJECTS

Coursework Deadline Optimiser | *Java, Spring Boot, Spring Security, JWT, PostgreSQL, JPA*

June 2026

- Built and deployed a full-stack coursework management application using Spring Boot, PostgreSQL and JWT authentication with a layered Controller–Service–Repository architecture.
- Engineered a scheduling algorithm that allocates study time across 60 days by calculating priority weights based on deadline urgency, task difficulty and course weighting
- Designed a relational database schema with complex JPA entity relationships and implemented full CRUD operations for all entities
- Configured a CI/CD deployment pipeline using GitHub and Render, automatically building, testing and redeploying the application on every push to the repository.

Board Game Management System | *Java, Git, CRUD*

May 2026

- Developed an end-to-end e-commerce system with inventory management, multi-method payments and role-based CLI interfaces for administrators and customers
- Implemented modular OOP architecture across 5 packages with polymorphic inheritance and role based access controls
- Developed multi-criteria inventory search with dynamic price sorting and real-time stock management

Loughborough University Route Finder | *Python, A* Search, Matplotlib, NetworkX*

May 2026

- Implemented A* pathfinding algorithm with Euclidean heuristic for optimised route computation across campus with obstacle avoidance and multi-waypoint support
- Built graph visualisation system using NetworkX and Matplotlib; mapped nodes with coordinates and edge weights for route analysis
- Developed state-space search framework with frontier management, explored set tracking and redundant node elimination for optimal path discovery

TECHNICAL SKILLS

Languages: Java, Python, C, MySQL, SQLite, PostgreSQL, H2, HTML, CSS

Frameworks: JUnit, PyTest, Spring Boot, Spring Security, JPA/Hibernate, Apache Kafka

Developer Tools: Git, GitHub, Github Actions, Docker, Jupyter Notebook, Eclipse IDE, IntelliJ IDEA, PyCharm, Maven, Postman

Libraries: JDBC, NetworkX, Matplotlib, Pandas, Dash, JWT